

Response to the Editor

Carbon Footprint and Carbon-Aware Selection of Document-Oriented,
Relational, and Hybrid Object-Relational Storage for Image-Based
Time-Series Workloads in Green IoT

We thank the Editor for a constructive and precise assessment, and for the opportunity to revise. Every point has been addressed, and the manuscript is substantially stronger as a result. In particular, the request for stronger evidence of applicability prompted a new set of experiments against real payloads recorded from a face-recognition-based attendance system in operational use. These both confirmed the paper’s central conclusion and uncovered a quantitative correction we would otherwise have reported uncorrected. We are grateful for that.

Below, each of the Editor’s points is quoted, followed by our response and the location of the change. One point is answered with partial rebuttal, which we set out explicitly rather than leave implicit.

1. Positioning within the existing literature

“The discussion should better position the proposed framework within the existing scientific literature, demonstrating more convincingly how it advances beyond previous studies rather than primarily summarizing the experimental findings.”

Response. We accept this fully. On re-reading, the Discussion explained *mechanism* and then moved directly to threats to validity, without returning to the literature at all. Every citation in that section was to vendor or technical documentation rather than to scholarly work. The Editor’s characterisation was accurate.

We have added a new Discussion subsection, “**Positioning against prior work**”, which states what changes relative to prior studies rather than restating what we measured:

- **An accepted engine ordering is inverted.** Prior benchmarks on scalar time-series data report MongoDB ingesting faster than TimescaleDB. We reproduce that at low resolution ($> 6\times$ faster at 360p) and then show it does not survive its own premise: the ordering inverts between 1440p and 4K, and MongoDB becomes *infeasible* at 6K. What the scalar literature presents as a property of the engines is a property of the payload regime those benchmarks occupy.
- **The carbon-accounting literature supplies methods, not answers at this granularity.** Green Algorithms, the Software Carbon Intensity specification, and the embodied-carbon literature can attribute a footprint to a workload once its energy is known, but cannot indicate which architecture will consume that energy. We measure per architecture, per operation, and per resolution, and we note that a single aggregate figure per architecture, which those methods would ordinarily yield, would have averaged the crossover away and produced a spurious universal winner.
- **The output is a decision instrument, not a benchmark table.** The framework is now expressed as a transferable rule (see point 3) rather than a recommendation tied to our hardware.

We also state explicitly what we do *not* claim: that the specific crossover resolutions generalise. Table 1 has additionally been extended with a column recording what each prior study concludes and what changes in our payload regime, so the advance is visible in the comparison itself rather than asserted in prose.

2. Dialogue with recent publications in this journal

“The manuscript would also benefit from a stronger dialogue with recent publications in *Environment Systems and Decisions*, reinforcing its contribution to the journal’s body of knowledge.”

Response. Accepted. The submitted version cited no work from this journal. We have added a new Related Work subsection engaging six recent articles substantively. Each is used to sharpen a specific methodological choice, not cited in passing:

- A systematic review of multi-criteria decision making for material selection [1] identifies a recurring weakness in decision frameworks: criteria are often taken from the literature rather than measured in the adopting organisation’s own context. This directly motivates our choice to key the rule to a variable the adopter can measure, namely payload size, and to supply a procedure for locating the boundaries locally rather than a fixed recommendation.
- A life-cycle assessment of lightweight railway vehicles [2] establishes sensitivity analysis as what separates a robust conclusion from an artefact of assumed parameters. We identify our multi-region grid-intensity analysis as playing exactly that role, and show the ranking is invariant to the most uncertain parameter in any carbon accounting.
- A data-driven study of smart-home energy consumption [3] treats building instrumentation as the object of environmental analysis. We draw the distinction that such models account for the energy the building consumes, but not for the tier that stores what the instrumentation records, which we show can be reduced by roughly half through a storage decision alone.
- Two studies of IoT operational integrity, on self-healing through log analytics [4] and real-time risk at the edge [5], both presuppose a data tier that durably retains device output. We supply the missing environmental cost of that presupposition.
- A review of price-driven energy management and demand response [6] prompts a substantive point rather than a citation: because emissions are energy multiplied by grid intensity, temporal or geographic load shifting rescales every architecture equally and cannot change which one to select. The saving must come from the architecture itself.

3. Practical and scientific value, and applicability beyond our environment

“At present, the framework is derived almost entirely from the experimental environment developed by the authors, and its applicability beyond the evaluated hardware, software configuration, and workload characteristics remains insufficiently discussed.”

Response. This was the most substantive point, and we have addressed it in three ways: analytically, empirically with new experiments, and by separating claims that depend on measurement from those that do not. We also offer one partial rebuttal, stated at the end.

3a. An analytical model, so the rule transfers rather than being copied

We now derive the mechanism behind the crossover explicitly. Each architecture pays a *fixed* per-operation cost a and a *per-byte* cost b , so storing a frame of size s costs approximately $E(s) = a + bs$. Two architectures with different slopes cross exactly once, at $s^* = (a_2 - a_1) / (b_1 - b_2)$.

MongoDB has the smallest fixed cost and the largest per-byte cost; the hybrid is its mirror image; inline PostgreSQL lies between on both axes, which is why it is best at no operation and worst at none. We fit these constants to our measurements and report them. **The consequence is that an operator need not adopt our resolutions:** two insertion runs per architecture at two payload sizes yield a and b on their hardware, and the equation gives their crossover. Our Table 5 is thus one instantiation of the equation, and the equation is what generalises.

3b. New experiments: real camera payloads, a second camera, and a different host

We describe these in the manuscript as **additional experiments carried out for this revision**.

The submitted benchmark inserted one collage image for every row, so all 2,000 stored rows were byte-identical, which is reproducible but not what a camera produces. We therefore drew a corpus of real frames from a **face-recognition-based attendance system** in operational use: an installation that captures images from one networked camera and two USB cameras in order to identify people at entry. This is precisely the class of deployment the paper addresses, in that it retains image streams alongside timestamped metadata in a continuously running data tier. We take from it only the *storage* workload, namely the images its cameras produce at the resolutions and capture rate the installation uses. The recognition pipeline itself, its detection, embedding and matching stages and their accuracy and energy cost, remains outside the scope of the paper, which concerns the carbon cost of retaining images rather than of analysing them; we have made that boundary explicit in the manuscript so that the reported figures are not read as a whole-system footprint. Ten minutes of capture at 5 frames per second yielded 2,845–2,988 frames per camera; every frame was verified byte-distinct, and the benchmark was re-run with a different real image for every inserted row. Three results follow:

1. **The architectural conclusion holds.** MongoDB is the lowest-carbon architecture at every resolution from 360p to 1080p, as the synthetic sweep reports for that range, and the result replicates independently on a second camera.
2. **The synthetic payload is close to a worst case,** not a typical one: real frames are $3.1\times$ to $5.4\times$ smaller at equal resolution.
3. **The identical-row construction overstated MongoDB’s margin.** MongoDB’s storage amplification at 1080p rises from $0.10\times$ with identical rows to $0.92\times$ with distinct frames, because bucketed duplicates are removed almost entirely by compression while real frames are not. At matched payload size, inline PostgreSQL’s carbon penalty relative to MongoDB falls from $3.96\times$ to $1.11\times$.

We report this third finding prominently and against our own interest, because it is material: the *direction* of the published result is confirmed and its *magnitude* is not, and the crossover is expected to occur earlier than the 1440p–4K interval measured synthetically. We also state the caveats bounding the comparison, since real frames differ from the collage in size as well as in distinctness.

The validation additionally ran on a *different computer*: a four-core Core i7-1165G7 with 15.30 GB of memory, against the eight-core Core i9-11900H with 30.99 GB used for the controlled benchmark, with the MongoDB cache scaled proportionally from 30 GB to 14 GB. This is therefore a second-hardware replication and not merely a second-payload one. Absolute figures differ between the machines, as they must; the architectural ordering does not. A new table in the manuscript specifies both environments side by side. We also state explicitly that the deployment supplied the cameras and the capture regime, while the storage measurements were executed on these controlled, fully specified hosts.

3c. Separating structural claims from measured ones

Two claims are now identified as independent of our hardware entirely: the 16 MB BSON ceiling is a protocol limit, so no hardware makes bucketed time-series collections able to store 6K frames; and the ordering of per-byte costs follows from where each architecture places the payload, not from how fast the host runs.

3d. Partial rebuttal: the single-host baseline is a design choice

Respectfully, we retain the single-host, single-client design and wish to justify rather than silently amend it. Multi-client concurrency changes how energy is attributed and introduces contention noise. It is effectively a different study, not a more realistic version of this one. The controlled baseline isolates payload handling, the variable under study, from scheduling and contention confounds, and keeps the comparison fully reproducible. We have strengthened the evidence for generality by the means above, which we believe answer the Editor’s concern more directly than added concurrency would, since the crossover is driven by payload size, a structural property of each storage design, rather than by client count. A multi-client evaluation remains identified as future work.

4. Limitations in the abstract

“The abstract should also explicitly acknowledge the main limitations of the research.”

Response. Accepted and implemented. The abstract now states, in the authors’ own words, that the study is deliberately bounded and names four limitations: the single-host, single-client controlled baseline rather than a distributed multi-tenant deployment; a limited photographic corpus, with real-frame validation reaching 1080p; measurement under one national grid intensity, with the explicit note that the *ranking* is grid-invariant because emissions scale linearly with that factor; and embodied-carbon and fleet-scale figures being first-order estimates from published factors rather than site-specific measurements.

5. Declaration of AI-assisted technologies

“If artificial intelligence tools were used for language polishing or translation, this should be transparently declared in accordance with the journal’s editorial policies.”

Response. Declared. A **Use of AI-assisted technologies** statement has been added to Statements and Declarations, recording that generative AI tools were used *solely for language editing*, that is grammar, clarity, and phrasing of author-written text, and that no AI tool was used to generate, analyse, or interpret the experimental data, produce figures or tables, or draw the scientific conclusions. The authors reviewed and edited all such output and take full responsibility for the entire content. AI is not credited as an author.

We took the opportunity to complete the declarations block, which was otherwise thin: Competing Interests, Data and Code Availability, and Ethics Approval have also been added. The ethics statement records that the new real-frame corpus involved no human participants and no personal data, was captured with no other person present, retained no biometric identifier, and that every saved frame was verified automatically to contain no detectable face before use.

Summary of changes

Editor’s point	Where addressed
1. Positioning in the literature	New Discussion subsection “Positioning against prior work”; Table 1 extended with a column stating what each prior study concludes and what changes here.
2. Dialogue with this journal	New Related Work subsection engaging six recent articles from this journal, each used to sharpen a specific methodological choice.
3. Applicability beyond our setup	New subsection deriving the mechanism behind the crossover and giving a measurement procedure for locating it elsewhere; new section reporting the real-case deployment, with a controlled comparison that separates payload construction from hardware; claims that do not depend on measurement identified separately.
4. Limitations in the abstract	Abstract revised to name four limitations explicitly.
5. AI declaration	Statements and Declarations rewritten, adding use of AI-assisted technologies, competing interests, data availability, and ethics approval.

References

All six appear in *Environment Systems and Decisions*. Every DOI was verified to resolve and the bibliographic details checked against Crossref before inclusion.

1. Bajwa AUR, Siriwardana C, Shahzad W, Naeem MA (2025) Material selection in the construction industry: a systematic literature review on multi-criteria decision making. *Environment Systems and Decisions* 45(1). 10.1007/s10669-025-10001-w
2. Kraft S (2023) Life cycle assessment in the development process of lightweight railway vehicles using sensitivity analysis. *Environment Systems and Decisions* 44(3):527–546. 10.1007/s10669-023-09955-6
3. Adams S, Greenspan S, Velez-Rojas M, Mankovski S, Beling PA (2019) Data-driven simulation for energy consumption estimation in a smart home. *Environment Systems and Decisions* 39(3):281–294. 10.1007/s10669-019-09727-1
4. Singh P, Saman Azari M, Vitale F, Flammini F, Mazzocca N, Caporuscio M (2022) Using log analytics and process mining to enable self-healing in the Internet of Things. *Environment Systems and Decisions* 42(2):234–250. 10.1007/s10669-022-09859-x
5. Radanliev P, De Roure D, Van Kleek M, Ani U, Burnap P, Anthi E (2020) Dynamic real-time risk analytics of uncontrollable states in complex internet of things systems: cyber risk at the edge. *Environment Systems and Decisions* 41(2):236–247. 10.1007/s10669-020-09792-x
6. Al Essa MJM (2025) A review on price-driven energy management systems and demand response programs in smart grids. *Environment Systems and Decisions* 45(1). 10.1007/s10669-024-09998-

We hope the revised manuscript now meets the standard required to proceed to an Associate Editor, and we thank the Editor again for guidance that materially improved the work.